CS231 Fundamental Algorithms
Section 01 (Brian's section)
Spring 2026 Syllabus

This course is about learning the algorithmic problem-solving strategies used by computer scientists to design and analyze algorithms. You can think of it as a continuation of CS230 and MATH225. CS230 helped you become a competent programmer: You learned to translate an algorithm into code using good programming practices (testing, modularity, abstract data types, etc.). You also learned about some basic data structures and began to think about time and space complexity. MATH225 taught you basic proof techniques, combinatorics, and graph theory. CS 231 will synthesize these experiences into the skills needed to design provably correct and efficient algorithms. Through the study of many classic example algorithms, we will learn high-level problem-solving techniques (greedy algorithms, dynamic programming etc.) along with tricks for proving properties of algorithms (running time, correctness, etc.).

Learning goals for the course

This course has been designed to achieve the following learning outcomes:

1. *Foundational Knowledge*: You will recognize, understand, and develop intuition for algorithmic problem-solving strategies that are considered to be fundamental knowledge for any computer scientist.
2. *Technical Knowledge*: You will learn how to analyze the complexity of the various solutions you may design or encounter and become fluent in expressing your thoughts and arguments about the efficiency of those solutions.
3. *Connect Content to Real-World Situations*: You will recognize real-world problems that could be solved with the algorithms or problem-solving strategies we study, and be able to interpret the real-world implications of their solution.
4. *Study and communication skills:* You will become more comfortable reading, speaking, and writing about challenging subject matter in the area of algorithms and theoretical computer science.

Instructors:

- Section 01: Brian Brubach https://bbrubach.github.io/

Tutors:

- Han Nguyen
- Samiksha Singh
- Karen Xiao

Prerequisites

CS230-Data Structures and *MATH225 -Combinatorics and Graph Theory* (or equivalent). If you do not have **both** of the pre-requisites, you need the instructor's permission to attend the class.

Textbook (required)

Algorithm Design by Jon Kleinberg and Éva Tardos. Pearson, 2006, ISBN 0-321-57351-X (at Wellesley's Bookshop). There is only one edition of this book. So used copies are fine.

Assigned reading should be done before class on the date listed on the class schedule. You don't have to understand everything in the reading before attending class. Your goal in the first pass should be to understand the main ideas and examples of the topic we will be covering. This will help you retain more information during the lecture, which helps even further when you do a second reading, but getting a little familiar with the material before the lecture is helpful. Assigned readings from "DPV" refer to the free DPV textbook provided in the resources folder of the student drive.

Helpful links and additional resources

- There are many visualizations of the algorithms we will study on David Galles' website.
- This is a nice reference for making math symbols in latex.
- Here is a long list of mathematical notation that you may have forgotten.
- Other resources are provided in the Resources folder on the shared google drive.
- These video lectures on many of the topics covered in our course. I generally discourage watching videos on the lecture content, but if you feel you must watch videos on this content, these are more trustworthy than random Youtube videos.
- Students in this course are expected to follow the CS Department Guidelines.
- De-texify

Course Communication

Our main modes of communication will be through google group emails, resources provided in this CS231-Students google drive folder, and drop-in hours. Please check the course schedule document regularly.

Drop-in hours schedule

All sections of CS 231 will share drop-in hours since we are covering the same material and using the same assignments. Thus, you are welcome to attend drop-in hours with either professor or any of the tutors.

- Brian Brubach:
    - **Mondays 11:30am-12:30pm** in H405
    - **Wednesdays 10:30-11:30am** in H305
    - **Thursdays 3:45-4:30pm** in H401
    - Or by appointment.
    - You are also welcome to visit my office (SCI W424) anytime, but questions about assignments should be limited to my drop-in hours or scheduled appointments.
- Tutor times/places:
    - **Sundays 4-6pm**, in H401 (Samiksha)
    - **Tuesdays 7-9pm**, in H401 (Karen)
    - **Thursdays 5-7pm**, in H401 (Han)

Hot tip: During weeks with no assignment due, drop-in hours are typically under-attended. That's a great time to come get extra help with any material you're finding difficult.

Guidelines for drop-in hours

Drop-in hours for this course can be challenging. Below are some guidelines to help things go as smoothly as possible.

- If you're stuck on a problem, come prepared with specific questions and try to identify where you might be confused about related content that was covered in the lectures.
- Do not ask instructors or tutors to look at what you've written and verify if it's correct. Sometimes it will be necessary for you to talk through your current solution to help us understand where you're getting stuck, but we cannot tell you whether or not your solution is "correct" in drop-in hours.
- For assignment problems, you can discuss high-level ideas or strategies with course tutors, but not detailed solutions.
- Please limit personal electronic device use while attending drop-in hours and give your instructor/tutor your full attention when asking questions.
- This is mostly a proof-based course, but we will have a few coding problems. For these coding problems, tutor drop-in hours should not be used for basic debugging questions.
- Most importantly, student tutors should be treated with respect at all times:
    - If a tutor says they cannot answer a question related to an assignment, do not pressure them.
    - When drop-in hours are busy, please accept that a tutor may only have a limited time to meet with you and may need to attend to other students.
    - Do not ask tutors to stay beyond their allotted time drop-in time.
    - Do not approach tutors outside of drop-in hours to ask about the course.
    - Remember that tutors may be just as busy and stressed out as you are.
    - Remember that tutors are human and may make mistakes.
    - If you're having an issue with a tutor or observe anything problematic during tutor drop-in hours, you can always reach out to either instructor to discuss it.

Mode of Instruction

We will meet:

- **Monday** from **9:55am-11:10am** in **SCI H103**.
- **Wednesday** from **9:30am-10:20am** in **SCI H103**.
- **Thursday** from **9:55am-11:10am** in **SCI H103**.

Participation in the class meetings is important. You will receive credit for your active participation. Frequent questions and especially **wrong answers are strongly encouraged**.

We will use double-sided name tents that are randomly distributed for each lecture. The side facing the instructor specifies whether you are okay with cold calling (calling on you to answer questions even when your hand isn't raised). Please sit at your name tent and choose at any time which way to face it. The name tents also let us randomize groupings for in-class exercises. If you need to sit in a particular area for any reason, please let the instructor know (e.g., sitting in the front because you forgot your glasses or sitting in the back after a concussion).

Wednesday classes

The Wednesday meeting times will mostly be used to practice and review what you've learned through group exercises rather than presenting new material. The hope is that we can use this time to explore the material in a less stressful and more social way than we do with the graded assignments/quizzes/exams. We will also use Wednesdays for our two mini midterm exams and to make up any lecture that gets cancelled or finish lectures that run over.

Electronic device policy

- Using tablets for note-taking is allowed. All notifications should be turned off (not just silenced).
- Laptop use in class is not allowed unless you have discussed accommodations with me. If you need to use a laptop for note-taking, please reach out to me ASAP to discuss this accommodation. If using a laptop, only your notes should be open, and all notifications should be turned off (not just silenced).
- Phone use in class is not allowed. Phones should be silenced and put away/not visible.
- Of course, if you need to leave class to take an urgent phone call, that's completely fine.

This policy is designed to create a focused, engaged, present, and professional learning environment. We should all be together in one space, separate from the rest of the world, and concentrating on learning with each other. We will give reminders of this policy before and during class as needed, but repeated failure to follow the policy will result in a deduction from your final course grade.

We also recommend putting your devices away before class begins and getting to know the people at your table who you will be working with on in-class activities, but this is not required.

Attendance and late arrival policy

- Attendance is required for **all class sessions (Monday, Wednesday, and Thursday)**.
- You are **allowed 4 unexcused absences on days without an exam**. This is intended to account for illnesses and other unavoidable absences that occur during the semester. You do not have to tell me why, but you should email by the start of class time to let me know. You are responsible for getting notes from another student to make up for missed content.
- **More than 4 unexcused absences will affect your final grade** in the course.
- If you need to miss more than 4 classes or need to miss an exam, please contact me to discuss a plan for keeping up with the content or taking an incomplete if necessary.
- Attendance includes arriving on time to participate in lecture and group exercises. **Every 2 late arrivals will be counted as an unexcused absence**.
- Note that our final exam is during exam week and flying home early is not an acceptable excuse for missing the exam.

If you struggle with attendance or being on time, I'm happy to meet with you to discuss a plan for improvement. One simple strategy for being on time is to give yourself a reason to arrive early.

In-class group exercises

We will have a number of in-class group exercises with random group assignments. The goal of these exercises is for everyone in your group to understand the content better or to discover what they have questions about. To this end, we have the following guidelines to make group exercises inclusive, supportive, and productive:

- Make sure everyone gets a chance to talk.
- When working at the white board, take turns writing.
- Take time to explain if one group member doesn't understand rather than racing ahead without them.
- Speak in a shared language that all of your group members understand.

Assignments

The goal of assignments is to reinforce the material we cover in class and practice transferring that knowledge to slightly different problems.

There will be six problem sets. Each problem set will have three parts:

1) **Sample problems** with example solutions. These will give you some idea of how you are expected to compose solutions to the proof-based problems in this course.
2) **Collaboration problems** where you are encouraged to discuss solutions with one or two other students. However, you must still write your solution by yourself.
3) **Individual problems** which you should not discuss with any other students.

Submission

Assignments must be **typeset using Latex** and submitted in pdf format to Gradescope. We recommend using Overleaf, especially if you're new to Latex, but you must **disable the Writeful tool** to comply with the AI policy. Additional typesetting and submission instructions will be on the first page of the assignment. All assignments are **due by 11:59pm (eastern time) on the due date** on the course schedule. The assignments may require some time for thought, and I strongly recommend you **start early**.

Late policy

Turning in assignments on time is crucial for the course to function efficiently and for us to be able to grade and return assignments as quickly as possible.

If you need more time to complete an assignment for any reason, you can take a free **48-hour extension** on **up to two assignments** by default, no need to ask for permission or email me about it. This applies **only to assignments, not exams or quizzes**. It is **your responsibility to keep track of how many late submissions you've made** and avoid using more than two. Beyond these two assignment extensions, all deadlines are hard deadlines, and any late assignments will be accepted at the instructor's discretion for reduced credit.

If circumstances arise in which you need an exam/quiz extension or any additional extension on an assignment (beyond 48 hours on two assignments), please contact me ASAP. For extensions other than the two free 48-hour extensions, you must contact me about the extension prior to the due date to discuss your options. If you're struggling to meet course deadlines, please reach out to me as early as possible, and I'll do my best to support you however I can.

Collaboration

For individual assignment problems and exam/quiz problems, there is no collaboration allowed at all. If you have questions about these problems, please contact your instructor or tutors as detailed in Course Communication.

For collaboration problems on assignments, you are allowed to discuss the problem with **other students** in the class and strongly encouraged to do so. However, you should **write your solution completely by yourself** and you **should not share your written solution with anyone ever**. You may not collaborate or discuss assignments with anyone outside of our class, especially with students who have previously taken the course.

General guidelines for collaboration:

- Working in groups of 2 or 3 is ideal.
- Inviting someone working alone to join your group is always encouraged, regardless of group size.
- Groups of 5 or more should consider splitting into smaller groups and instructors/tutors may ask large groups in drop-in rooms to split up.
- Speak in a shared language and make efforts to be inclusive of all group members. If someone is shy or not speaking much, pause to ask for their thoughts or input.
- Instead of solely focusing on solving assignment problems, take time to review the lecture content which will give you the foundation you need before beginning the assignment problems.
- No taking pictures of whiteboards or other students' notes or sharing such pictures.
- Avoid listening in on other groups or studying what they write on whiteboards.
- Erase whiteboards when your group is finished using them.

These guidelines are designed to help you maximize your learning from assignments and avoid temptations to take shortcuts just to complete an assignment.

Internet resources

You are **not allowed to search for answers to any assignment or exam problems online**. This includes, for example, googling text or keywords from the problem description. Use of internet resources to understand basic concepts covered in the course is permitted, but discouraged as there's a lot of misleading or incorrect information about our content. However, for the coding problems, **using internet resources for basic coding stuff is totally fine** (e.g., Googling "How do I make a list of lists in python?" is okay), but **you should not be searching for or viewing code that solves the assigned problem**.

Beware of youtube videos from unknown sources that sometimes give inaccurate information. If you're looking for videos covering the topics we're studying, I recommend the video lectures on this page which cover most of the topics we'll be studying. However, **I strongly advise you to use our class textbook as your primary study guide since reading an algorithms textbook is a valuable skill to develop**. If you rely exclusively on class lectures, slides, and online videos, you will miss out on developing that skill.

AI policy

No use of AI/generative AI is permitted in this course. You may not use AI to compose, edit, provide feedback on, or engage in any way with any content in this course. This includes both proof-based and coding problems. Use of AI on any submitted material will constitute cheating and may be subject to an honor code charge. In addition, uploading assignment problems or other intellectual property associated with this course to an AI service (or any other entity) will constitute a separate serious violation of the honor code. As a simple rule, anything you submit should be something you could have written with only a pencil and a piece of paper.

While AI tools can be helpful in completing certain tasks, AI-assisted writing conflicts with the learning goals of this course and is a separate skill which will not be taught in this course.

Note: This means Overleaf users should disable the Writefull tool when working on assignments for this course. Writefull appears as yellow underlines with suggested text. You can disable it by clicking “disable” in the pop up.

Rules for after the semester

- Some students may wish to include work from CS 231 in portfolios, resumes, etc. Please do not post CS231 work publicly (including on GitHub, GitLab, Bitbucket, your own website, etc.), but feel free to show your work privately to potential employers.
- Do not make your work accessible or viewable by current or future CS 231 students.

Quizzes

We will have 6 Gradescope quizzes released roughly every other week and **due two days after being posted at 11:59pm** Some quizzes may be moved to different days due to holidays and this will be noted on the course schedule and in Gradescope.

Quizzes will always be **individual assignments**, but open-book. So you **may not talk to others about the quiz questions or seek answers online**. However, you may use the textbook, your notes, and other course resources.

One purpose of the quizzes is to give you frequent, immediate feedback. So the quizzes will be mostly auto-graded with results released soon after they are collected. Another purpose is to give you more graded content that is “easier” than the assignment problems, but still demonstrates an understanding of the material. This takes some of the grading weight off of assignments and exams.

To accomplish these goals, questions on each topic will usually be spread across two quizzes. When a given topic appears on a quiz for the first time, the questions will generally be easier and some of them may even be worth zero points. These questions will help you discover your misunderstandings before the homework problem set is due for that topic. When the topic appears again on the next quiz, the questions will be a little harder and may sometimes be worth more points.

Two more things to note about quizzes:

- I will **drop your lowest quiz grade**.
- I will set up gradescope to accept quizzes up to 24 hours after the due date. However, **accepting late quizzes will be entirely at my discretion**. This just means that if your computer crashes while you're taking the quiz five minutes before the deadline or you forget to take a quiz, you can still submit something up to 24 hours after the due date without having to email me about reopening the quiz.

Midsemester presentation

Each student will perform a midsemester presentation.

- This will be a one-on-one presentation to the instructor outside of lecture time.
- The presentation should be 15-20 minutes plus about 5 minutes of Q&A from the instructor.
- More details about the presentation topic, format, and evaluation will be provided in a separate document later in the semester.

Mini Midterm Exams

There will be two short midterm exams. These will take place during our Wednesday meeting times. The goals of these exams are to:

- Give you additional opportunities to demonstrate your knowledge.
- Allow the more challenging take-home assignments to be a smaller fraction of your final grade.
- Give you practice taking a pencil and paper algorithms exam before the final exam.

Final Exam

We will have an **in-person final exam during finals week**. The exact date and time will be announced when we are given our scheduled time by the college.

Other exam details:

- The exam will be comprehensive.
- The exam will be **closed book**, but you can bring a cheat sheet. The cheat sheet must be **one double-sided piece of 8.5”x11” paper with your name and id printed clearly at the top and must be turned in with your exam**.
- The exam problems will be “easier” than the assignment problems and some may be very close to what we have covered in a lecture.
- There will not be a practice exam, but I may offer some practice problem examples.
- Typical exam rules will be in place, including **no electronic devices and no discussing the exam even after you have completed it**.
- I generally do not curve exams or any other graded material. However, I will adjust the final exam slightly if necessary.

Reading checks

During the **first four weeks** of the course, there will be a mini Gradescope reading check due before each lecture (not before Wednesday sessions).

- Each check is **due by 9 AM on the day of the lecture**.
- You are allowed to consult any of the assigned readings to answer the questions, but there is **no collaboration or use of internet/AI resources allowed**.
- They are **not timed** and you can open them to look at while you do the reading if you want.
- There will be 7 of these (no check before the first lecture) and each check will be five questions worth one point each.
- There are 35 total points you can earn, but **your final grade for reading checks will be out of a maximum of 20 points**. In other words, there is no benefit to earning more than 20 points, but earning fewer than 20 points will have a tiny effect on your grade.
- There will be **no late submissions or makeups for reading checks** except in extreme circumstances. Note that you can technically get all 20 points by only doing 4 of them.
- The goal of this is to give you a low-stress nudge to complete the reading before class and give me some feedback before lecture on any parts that folks didn’t understand.

Evaluation

Grading

- ***Class attendance, participation, preparation, reading checks*: 10%** (This is distributed among many small things such as the reading checks and completing "Assignment 0". I expect everyone who does what is required for the course to get full credit here)
- ***Assignments*: 36%** (6 assignments, 6% for each assignment)
- ***Quizzes*: 10%** (6 quizzes, lowest quiz grade is dropped and 2% each for the other 5 quizzes)
- **Mid-semester oral presentation: 10%**
- **Mini Midterm Exam 1: 4%**
- **Mini Midterm Exam 2: 5%**
- **Final Exam: 25%**

The Honor Code

The course collaboration policy is outlined above. Violation of this policy will constitute a violation of the honor code. **Violations include, but are not limited to: sharing any of your written solutions, viewing any other student's written solutions, submitting work that is not your own, searching for solutions online, and any collaboration or discussion of exam problems or individual assignment problems**. Any infractions will be reported to the Honor Code Council. The reason for this policy is to help you learn by thinking actively about the topics of the course. Adherence to the honor code also gives us greater flexibility to have things like self-scheduled exams.

Note that sharing any of your solutions to assignments, quizzes, exams, etc. with another student for any reason is a violation of the honor code **even after you have completed the course**.

Disability-Related Accommodations

I welcome your inquiries regarding disability-related accommodations you might need to take this course. Please first read through the syllabus to get a sense of what is offered and how. You are welcome to talk with me regarding your needs, especially if they relate to the course format and your learning styles and strengths. Be aware that for most accommodations, such as extended time on exams and notetakers, you will need to discuss these with Accessibility and

Disability Resources (ADR) staff and they will need to notify me in writing of your approved accommodations.

If you find you will need academic accommodations, please look at the ADR website at wellesley.edu/adr. First-time users of their services will be asked to fill out an online application and upload documentation of their disability, which may be related to health, mental health, or learning disabilities and may be temporary or permanent in nature. You can find this application at https://shasta.accessiblelearning.com/wellesley and should use your Wellesley ID to log in. A staff member from ADR will follow up with you to discuss your requests and needs. If you have questions before applying, feel free to contact accessibility@wellesley.edu.
If you have already been approved for services through ADR, you may go directly to the online accommodation management program (https://shasta.accessiblelearning.com/wellesley) and choose which of your eligible accommodations you would like included in a notification letter that will be sent to me.

Please submit your accommodation requests within the 1st week of the semester or as early as possible so that ADR and I can address your requests. ADR is not able to guarantee they can address your request if it comes after the deadline but will do the best they can.

Ombuds office

The Ombuds Office is a confidential and neutral resource for students who want to discuss various options for dealing with issues such as interpersonal conflict in their campus lives: ombuds@wellesley.edu; x3385.

Peer support resources

Need support? Get to know the different peer resources available on campus!

Who can help me with my...

Drop by the PLTC on the 3rd floor of Clapp Library to learn more about Wellesley's academic success resources.

Organizational Skills?

Academic Success Coach (ASC)

ASCs are accountability partners trained to assist their peers in a number of areas necessary for student success, including registration, time management, and study skills. Use TracCloud for drop-in hours and appointments. Follow the ASCs on Instagram! @ascswellesley

Public Speaking?

Public Speaking Tutors

Public Speaking Tutors are available to meet with both one-on-one and in groups to offer you support in giving presentations, speaking at conferences, or looking to improve your vocal presence.

Writing?

Peer Writing Tutors

Tutors provide one-on-one guidance on any piece of writing you're working on, at any stage of the writing process. The tutors' goal is to help you generate, organize, revise, and improve your writing.

Career?

Peer Career Tutors (PCM)

PCMs support you as you prepare resumes, cover letters, apply to internships, and more. They host virtual drop-in and in-person cultural house drop-in sessions. Dates are posted on Handshake.

Health Questions?

Sexual Health Educators (SHE)

SHEs are peer educators who work with the Wellesley College Office of Student Wellness to provide medically-accurate and inclusive education and outreach in the area of sexual health. Follow the SHEs on Instagram! @wellesleyshes

Course Material?

Peer Subject Tutors

Tutors provide subject-specific tutoring across the curriculum. They feature “café” style drop-in help for many courses, including some course and section-specific “office hours,” as well as individual tutoring in selected courses. Schedule an appointment on TracCloud.